Research Statement

I am an applied economist working in industrial organization and the economics of innovation. My research asks how and why firms allocate their research: the mode of that research, why they enter certain areas, and how government assistance and changes in market structure affect the scale and composition of their portfolios. I combine applied theory with applied econometrics, using production and innovation surveys, patents, award records, and merger files.

Internal and External R&D: An analysis of costs and benefits. I study the costs and benefits of in-house and contracted-out R&D among Dutch ICT firms from 2000 to 2020. A dynamic model finds that productivity gains are similar across the two activities, while contracting R&D out is much more costly than doing it in-house, which is why so few firms only contract out. For a firm that also does in-house R&D, 94 percent of that extra cost disappears. I interpret the gap as a hold-up problem in the external contract. A uniform subsidy raises R&D participation and firm value by more than a subsidy for in-house R&D only.

Entry into Innovation Areas. I then ask where firms put that research. The paper studies entry by U.S. electronics firms across innovation areas. Topic modeling on more than 630,000 patents defines those areas from the patent text, and I merge them with market-implied patent valuations. Firms occupy several areas at once, and the number of firms differs widely across areas. An entry game, estimated by moment inequalities, finds that additional rivals reduce expected profit and that firms gain from occupying more areas at once. In the semiconductor case, a payment based on a quarter of predicted value shifts 0.1–0.5 percent into those areas and leads up to 0.8 percent of firms to add an extra area. A 25 percent cut in the cost of occupying an area shifts up to 7 percent into those areas and leads up to 10 percent of firms to add an extra area.

The Impact of Government Financial Assistance on the Scope and Direction of Firm Innovation, with Costanza Cincotta. Governments spend hundreds of billions of dollars each year financing private innovation. We ask whether this spending expands, redirects, or reallocates research. We take a Salop model to U.S. public firms from 2000 to 2021, comparing recipients with close and distant rivals using Good Jobs First awards, patent data, and text-based product-market overlap. Distant rivals provide the preferred reading of the treated firm's own response. Recipients' patenting scale rises by about 4 percent relative to distant product-market controls. Nearby rivals do not change how much they patent. Direction shifts toward the industry's most-cited technologies, and the shift is strongest for cash grants.

Mergers and Innovation: An Exploration of Scope and Direction of Innovation, with Sabien Dobbelaere and José Luis Moraga-González. How do mergers change the scale and composition of firm innovation? We study U.S. public–public mergers from 1980 to 2020 using combined acquirer–target patent portfolios. A compact framework isolates business stealing, knowledge pooling, and activation, and keeps product-market overlap distinct from technological overlap. Comparing treated merger pairs to matched control pairs, joint-portfolio scope is larger after mergers, by about 0.18 on the annual measure. The difference is larger when pre-merger technologies overlap, especially among non-horizontal deals. Joint portfolios also show higher alignment with highly cited patents.

These papers treat the allocation of research as a set of discrete choices over modes, areas, and portfolios. Costs, rivals, cash, and ownership then move both how much firms invent and what they invent.

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